Project 2:-Maven Project 2

Step 1: Create a Maven Project:

mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-arch
etype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/test/MyMavenGuavaApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.549 s
[INFO] Finished at: 2026-03-17T20:02:50+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test$
```

STEP 2: Navigate to the Project Directory using cd MyMavenGuavaApp

```
harith@harith-VirtualBox:~/devops/test/MyMavenGuavaApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/harith/devops/test/MyMavenGuavaApp/target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.296 s
[INFO] Finished at: 2026-03-17T20:06:46+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenGuavaApp$
```

Step 3: Edit pom.xml: gedit pom.xml

pom.xml
~/MyMavenGuavaApp

```xml
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>

<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>


<plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.1.0</version>
            <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>
        </plugin>

</plugins>
</build>
</project>
```

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Step 4: Modify App.java

gedit App.java

App.java
~/MyMavenGuavaApp/src/main/java/com/example

```java
package com.example;

/**
 * Hello world!
 *
 */
// Import FileUtils class from Apache Commons IO library
// This is used to perform file operations like copying files
import org.apache.commons.io.FileUtils;

// Import File class to represent file and directory pathnames
import java.io.File;

// Import IOException to handle input-output errors
import java.io.IOException;

// Import ImmutableList from Google Guava library
// ImmutableList creates a list that cannot be modified after creation
import com.google.common.collect.ImmutableList;

// Define the main class
public class App

    // Main method - program execution starts from here
    public static void main( String[] args )
    {
        // Create an immutable list of strings (fruits)
        // This list cannot be changed (no add/remove allowed)
        ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");

        // Print the list of fruits to the console
        System.out.println(fruits);

        // Create a File object representing the source file
        File sourceFile = new File("source.txt");

        // Create a File object representing the destination file
        File destFile = new File("destination.txt");

```

Step 5: Build the Project using the command mvn compile command

```
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/harith/devops/test/MyMavenGuavaApp/target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.893 s
[INFO] Finished at: 2026-03-17T20:22:24+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenGuavaApp$
```

Step 6: Run the Application:

If java-jar command give an error which is most probable,

use the following command : mvn exec:java -Dexec.mainClass=”com.example.App”

```
harith@harith-VirtualBox:~/devops/test/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MyMavenGuavaApp >---------------------
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
Error occurred while copying file: File system element for parameter 'source' does not exist: 'source.txt'
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.776 s
[INFO] Finished at: 2026-03-17T20:24:40+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenGuavaApp$
```

This command gives error because source.txt doesn’t exist.

Now, Create a Sample File with some content.

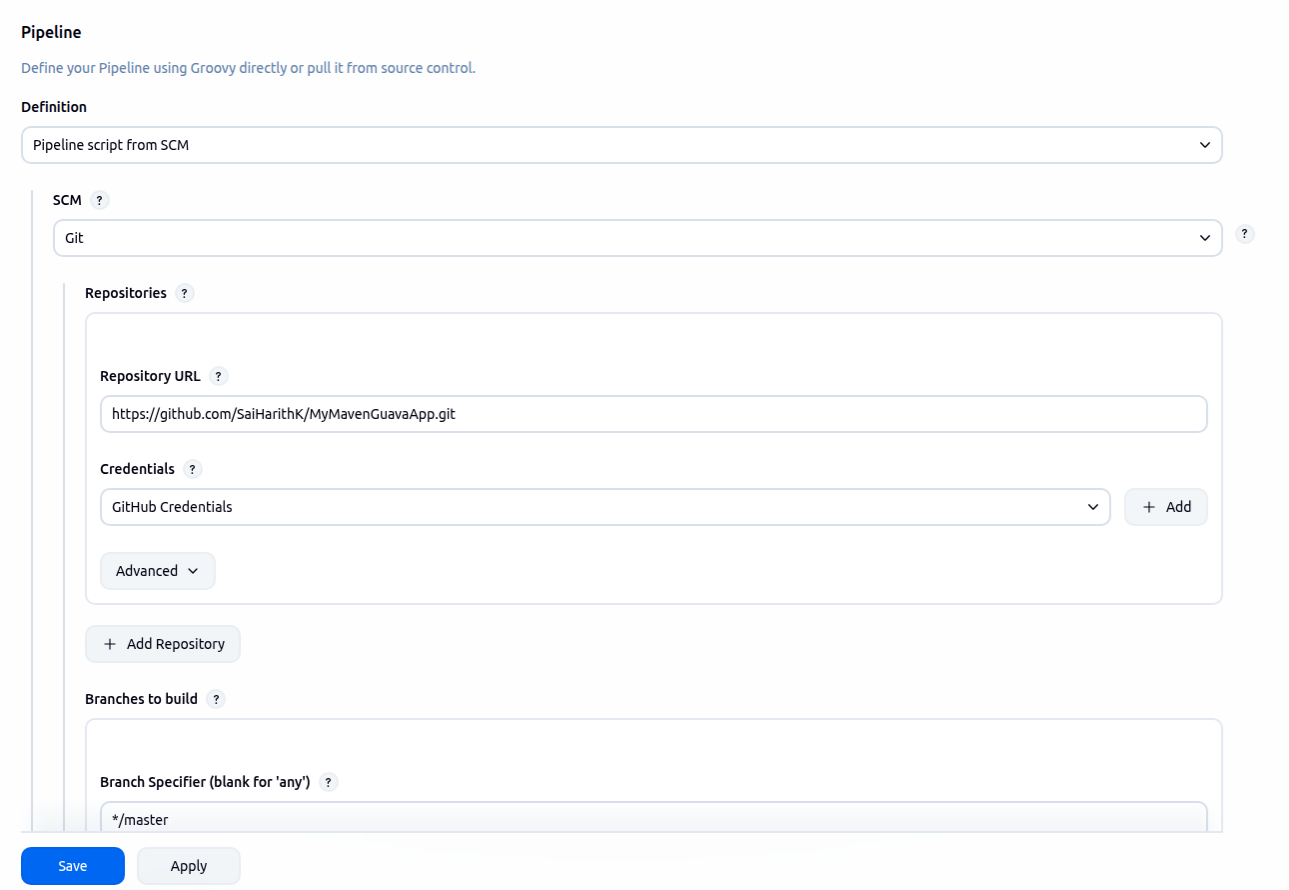
Run the command mvn exec:java -Dexec.mainClass=”com.example.App” again

The output shows file copied successfully.

Step 7: Pipeline

-login to jenkins account

-create a new item and configure the following settings



Step 8: Edit Jenkins file

-Create a Jenkinsfile and add the following code:

```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/SaiHarithK/MyMavenApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
```

```
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 9: Build

Click on build now and check if its successfull

MyMavenGuavaApp

Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Average stage times: (full run time: ~33s)	1s	323ms	2s	8s	4s	998ms	191ms
#1 Apr 13 18:58 No Changes	1s	323ms	2s	8s	4s	998ms	191ms

Permalinks